

A highly antifouling and eco-friendly hydrogel coating based on capsaicin derivative -functionalized polymer

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ABSTRACT

Antifouling hydrogels have received extensive attention in the field of marine antifouling due to their excellent hydrophilicity and environmental friendliness. However, the antifouling applications of traditional hydrogels are limited by their poor antibacterial properties, and capsaicin derivatives, as nontoxic and environmentally friendly antifoulants, can be used to improve the antibacterial properties of hydrogels. From the perspective of practical application, hydrogels were obtained by crosslinking the copolymer of the capsaicin derivative N,N'-((4,5,6-trihydroxy-1,3-phenylene) bis (methylene)) diacrylamide (TPA), lipophilic monomers and hydrophilic monomers. This hydrogel exhibited anti-protein performance due to its hydrophilicity, and the capsaicin derivative caused the hydrogel coating to exhibit excellent algae inhibition performance (the minimum adhesion density of diatoms on the surface of the hydrogel coating was less than 440 cells/mm²), better algae anti-adhesion abilities than hydrogel reported in the other paper, and antibacterial effects (the antibacterial rate was more than 95%). After 73 d of panel testing in a real marine environment, there was still a good antifouling effect in the hydrogel coating. Hence, the antifouling performance was improved significantly by capsaicin derivative (TPA)-functionalized hydrogel coating. A new strategy is provided in this study for the preparation of hydrogels with superior marine antifouling properties.

1. Introduction

Marine biofouling normally refers to the rapid adhesion and harmful accumulation of organic molecules and marine living matter on material surfaces immersed under seawater, which have caused serious economic and environmental problems (Gu et al., 2020). The biofouling process generally experiences three stages on the surface of the material as follows: 1) Initial adsorption and accumulation of some organics, such as protein and polysaccharide molecules; 2) bacteria, diatoms and other microorganisms adhered on the surface to form biofilms; 3) finally settlement and growth of large fouling organisms, such as mussels, tube-worms and macroalgae (Amara et al., 2018; Maan et al., 2020; Xie et al., 2011). Various studies have been developed to solve these problems. Among them, biocide-based antifouling coatings are the dominant

technology (Lindholdt et al., 2015), and most of the antifouling agents currently used are toxic and not environmentally friendly (Amara et al., 2018).

Hence, environmentally friendly hydrogel coatings that can prevent protein adsorption to prevent the occurrence of marine biofouling in the first stage through their surface hydration layer (Chen, 2022; Yang et al., 2021; Zhang et al., 2021) have received widespread attention. However, the antifouling application of hydrogels is still limited by their unsatisfactory antibacterial property (Zhang et al., 2022). Therefore, various approaches have been developed to improve the antibacterial properties and antifouling properties of hydrogel coatings, and the antifouling mechanism of biocide-based antifouling coatings used in hydrogel coatings is the most effective method and includes the introduction of silver nanoparticles (Zhou et al., 2022), copper compounds (Ashraf and

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Edwin, 2016; Lin et al., 2019), TiO_2 (Natarajan et al., 2018), and chitosan (Buzzacchera et al., 2017; Zhao et al., 2022). These materials, however, are toxic, result in environmental pollution and ecological hazards; or require expensive extraction, limited extraction technology, difficult mass production. Capsaicin is a promising marine antifoulant and an important component of environmentally friendly antifouling coatings. Capsaicin has come to be recognized as one of the most promising antifoulants, but its extraction process is complex, low rate, high cost and poor durability (Gu et al., 2020; Liu et al., 2019). Hence, many capsaicin derivatives with structures similar to that of capsaicin have been synthesized by simple and efficient methods (Wang et al., 2020, 2021). Moreover, these capsaicin derivatives have also been proven to be promising environmentally friendly marine antifoulants because of their excellent antibacterial and other antifouling activities against biofouling organisms (Tang et al., 2020; Wang et al., 2019). Among them, as a kind of antifouling capsaicin derivative, N,N' -((4,5,6-trihydroxy-1,3-phenylene) bis (methylene)) diacrylamide (TPA) could be synthesized simply and conveniently, and a good marine antifouling performance would be still exerted when TPA was doped into the resin physical (Cheng and Yu, 2010). In addition, the antifouling performance of the membrane can be significantly improved through the introduction of the capsaicin derivative N,N' -((4,5,6-trihydroxy-1,3-phenylene) bis (methylene)) diacrylamide (Zhang et al., 2022). These demonstrates that TPA is an environmentally friendly and effective antifouling agent, but no relevant investigation about its introduction into hydrogels for marine antifouling has been reported. Therefore, it can be expected to introduce into the hydrogel coatings by polymerization to improve marine antifouling performance.

Herein, a kind of capsaicin derivative N,N' -((4,5,6-trihydroxy-1,3-phenylene) bis (methylene)) diacrylamide, and other monomers were polymerized into hydrogels. The antifouling properties of hydrogel

materials were measured through anti-protein, algae antiadhesion, antibacterial experiments and real marine environment panel evaluation. The microscopic morphology, adhesion and other physical and chemical properties were investigated. This demonstrates that there is application potential for the adhesive ability of hydrogel coatings containing a capsaicin derivative, and the antifouling properties of hydrogel coatings were obviously improved with the introduction of a capsaicin derivative, more than 90% of the antibacterial rate and less than 1600 cells/ mm^2 of the number of attached *Nitzschia closterium* on the surface of hydrogel coatings. Therefore, the antifouling mechanism can be enriched of hydrogel coatings by introduced TPA capsaicin derivative and hydrogel coatings with good antifouling performance were successfully obtained, providing a new strategy for marine antifouling.

2. Materials and methods

2.1. Materials

Butyl acrylate (BA), azodiisobutyronitrile (AIBN), 2-(dimethylamino)ethyl methacrylate (DM), [2-(methacryloyloxy)ethyl]trimethylammonium chloride solution (DMC), acrylic acid (AA) and acrylamide (AM) were bought by Aladdin. Trifunctional azide crosslinking agent (TTMAP) was provided by Shanghai Chemical Reagents Institute. Pyrogallol, concentrated sulfuric acid (H_2SO_4), bovine albumin and anhydrous ethanol were purchased from Sinopharm Chemical Reagent Co. Ltd. Xylene and n-butanol were bought from Qingdao Datang Co., Ltd. 2216E agar and 2216E liquid medium were bought from Haibo Biotechnology Co., Ltd. SCDLP liquid medium and phosphate buffer solution (pH = 7.2) were purchased from Beijing Land Bridge Technology Co., Ltd. The capsaicin derivative N,N' -((4,5,6-trihydroxy-1,3-phenylene) bis (methylene)) diacrylamide (TPA) was synthesized in our

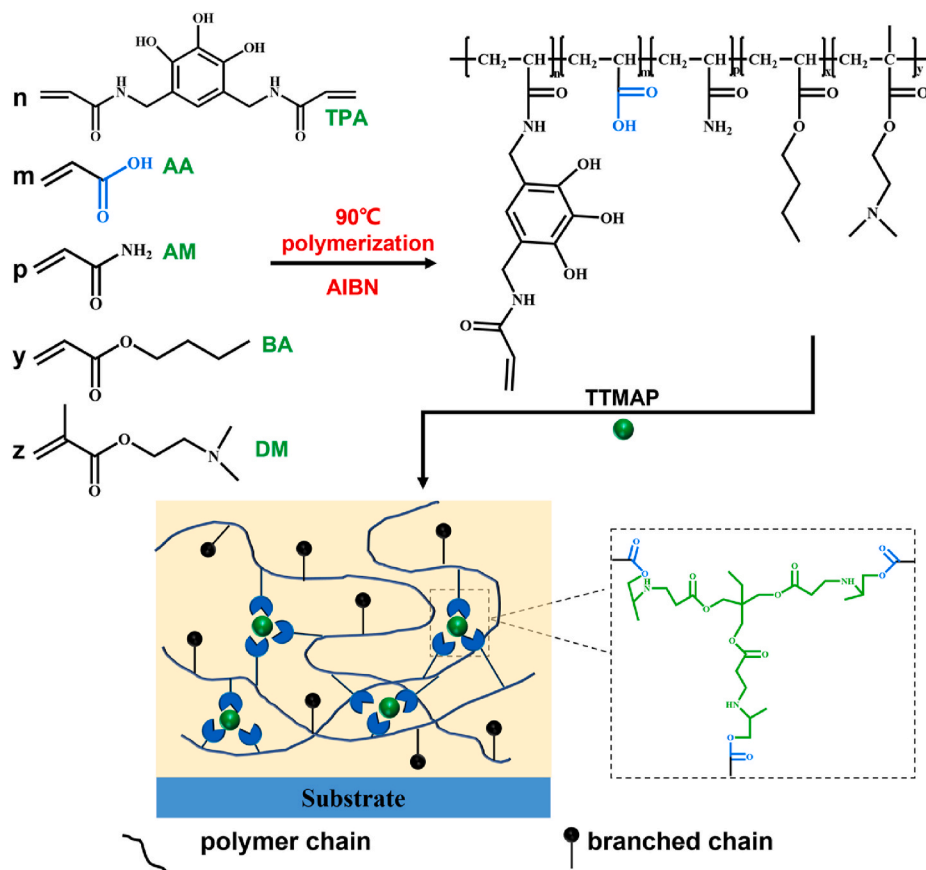


Fig. 1. The preparation process diagram of hydrogel prepolymer and hydrogel coating.

laboratory, its structure is showed in Figs. 1 and 2. Seawater was taken from a sea area in Qingdao. ABS and PVC were used as substrates for coating, both of which were provided by Qingdao Yixin Plastics Co., Ltd.

2.2. Preparation of hydrogel prepolymers

2.2.1. Hydrogel prepolymer containing DMC (PDMC5)

The monomer mixture (M1) was composed of 75 mL BA, 6.03 g AM, 10.5 mL AA and 7.2 mL DM, and M2 was 8 mL DMC. First, 1.2 g AIBN (initiator), 10 mL xylene and 10 mL n-butanol were mixed into the initiator solution (I1). The solvent (94.7 mL xylene, 45.5 mL n-butyl alcohol and 19.0 mL ethyl alcohol) was added to a four-necked round-bottomed flask equipped with a stirrer, a spherical reflux condenser tube and a thermometer, and then the flask was heated in an oil bath. When the temperature of the solvent reached 65 °C, 10 vol% of M1, M2 and I1 were placed into the flask, and the temperature was increased to 85 °C continuously, after the temperature was held for 40 min, M1, M2 and I1 were added, and the flask was heated to 90 °C. Additionally, starting from the second addition of the monomer and initiator, in addition to the first addition of M1 and I1, the remaining M1 and I1 were divided into three parts by volume, and each part was added at an interval of 20 min, and M2 was divided into six parts by volume, and each part was added at an interval of 10 min. One hour after the addition of M1, M2 and I1, 0.12 g AIBN was added, followed by maintenance for 3.5 h at 90 °C to obtain the prepolymer PDMC5 containing DMC. The temperature increase rate is 3 °C min⁻¹.

2.2.2. Hydrogel prepolymer containing TPA (PTPA3)

The monomer mixture (M3) was composed of 76.8 mL BA, 10.4 mL AA, 5.98 g AM, 7.1 mL DM, 7.38 g TPA, 48.8 mL xylene, 25.9 mL n-butyl alcohol and 8.9 mL ethyl alcohol. Five grams of AIBN (initiator), and 20 mL of xylene were mixed into the initiator solution (I2). 94 mL of xylene, 60.5 mL of n-butanol and 20.7 mL of ethyl alcohol were added to a four-necked round-bottomed flask equipped with a stirrer, a spherical reflux condenser tube and a thermometer, and then heated in an oil bath. When the temperature of the solvent reached 65 °C, 10 vol% of M3 and I2 were placed into the flask, and the temperature continued to increase to 85 °C. After the temperature was held for 40 min, M3 and I2 were added and heated to 90 °C. Additionally, starting from the second addition of monomer and initiator, in addition to the first addition of M3 and I2, the remaining M3 and I2 were divided into three parts by volume, and the addition interval of each part is 20 min. An hour after the all addition of M3 and I2, 0.5 g AIBN was added, followed by maintenance for 3.5 h at 90 °C to obtain the prepolymer PTPA3 containing TPA. The synthesis process of PTPA1, PTPA5 and PTPA7 is the same as that of PTPA3, their prepolymer composition is shown in Table 1, the synthesis process of the hydrogel coating is shown in Fig. 1.

2.3. Hydrogel coating preparation

At room temperature, the mixture of hydrogel prepolymer and crosslinking agent TTMAP was stirred for 5 min, after the mixture was static and defoaming, it was absorbed by a rubber head dropper and then evenly dripped onto the surface of the substrate pretreated with ABS by a drip coating method, dried for 48 h and further dried at 40 °C (Oven, DHC-9140A) for another 48 h. The crosslinking was completed and the solvent was volatilized completely. Finally, the hydrogel was obtained and weighed (w_s). Before the experiment, the ABS plate used was polished rough with sandpaper (No.80), carefully washed with water, desiccated and weighed (w_0). The size of the ABS panel for the experiment used were 50 × 50 × 1 mm³ (for the anti-protein, algae adsorption and antibacterial assay experiments), and 100 × 80 × 2 mm³ (for the static seawater immersion experiment).

2.4. Characterization of hydrogel coatings

2.4.1. Structural characterization

The structure of TPA was determined by nuclear magnetic resonance spectra (Bruker AVANCE III 600), TPA was dissolved in deuterated DMSO and tested in a nuclear magnetic tube by ¹H NMR. The molecular weight distribution of polymer was tested by gel permeation chromatography (Agilent infinity 1260 II), the hydrogel was dried and ground, and then dissolved in DMSO. The hydrogel prepolymers and their corresponding hydrogel coatings were dried and ground into powder, the same quality of KBr and sample was used to grind and tablet, and their molecular structures were determined by FTIR (BRUKER, EQUINOX55) at room temperature.

2.4.2. The characterization of hydrogel coating's surface

The surface morphology of the hydrogel coatings was observed by laser confocal microscopy (OLYMPUS, OLS4000), the ABS plate coated with hydrogel coating was placed under laser confocal observation. The fluorescence imaging of hydrogel coatings was tested by fluorescence microscope (Leica, CTR5000), the hydrogel coating was coated on a glass slide and placed under a fluorescence microscope for observation. Fluorescence excitation wavelength: blue light is 400–500 nm, green light is 460–500 nm, red light is 600–700 nm. The static contact angle of the hydrogel coating's surface was tested by a contact angle measuring instrument (KRÜSS, DSA 100) to determine its hydrophilic property and hydrophobicity. The prepared hydrogel coating sample plate was placed horizontally on the lifting platform, and 2 μL of deionized water droplet was injected at a speed of 2.0 μL per second. The droplet was fell onto the surface of hydrogel coating, and the contact angle was tested, three parallel points were selected to measure the average value to obtain the final contact angle. For the experiments in which the sample with

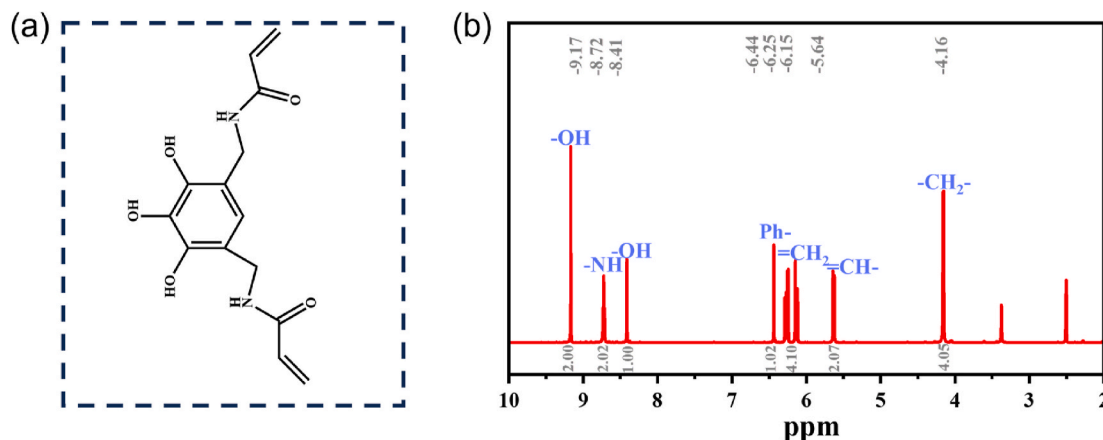


Fig. 2. (a) Molecular structure and (b) ¹H NMR result of TPA.

Table 1

The change in monomer composition and content in prepolymer (mol%).

Hydrogel	Prepolymer	Hydrophilic monomer	BA	AM	AA	DM	DMC	TPA
DMC5	PDMC5	38	62	10	18	5	5	
TPA1	PTPA1	34	66	10	18	5		1
TPA3	PTPA3	36	64	10	18	5		3
TPA5	PTPA5	38	62	10	18	5		5
TPA7	PTPA7	40	60	10	18	5		7

Note: the solid content of PDMC5 was 40%, and other prepolymers' solid content is 30%.

hydrogel needed to be removed from seawater and tested, the pre-treatment method of the hydrogel coating in the following: After the sample was taken from the seawater, the coating's surface was washed with deionized water, and then all the water on the surface was gently absorbed with absorbent filter paper.

2.5. Adhesive performance of the hydrogel coating

Hydrogel coating is different from the general coating, it has a certain elasticity after absorbing water, and there are some changes in its mechanical properties. Thus, general adhesion measurement methods such as the pull-off method, circle method and Baige knife method could not be conducted effectively. "Ship coatings—Determination of resistance to salt water—Salt water and hot salt water immersion method" (GB/T 10834-2008) was referenced. The "static seawater immersion panel experiment" method (Dong et al., 2022b) was used to measure the adhesive performance of the hydrogel coating to determine whether it had practical application potential. In the experimental stage, the prepared sample with hydrogel coating was placed in seawater, and the sample was completely immersed in seawater, 10 days was taken as an observing cycle, the adhesion performance of the hydrogel coating was compared by observing the time and area of the hydrogel coating falling off from the substrate.

2.6. Water-absorption test and weight-loss experiment

The antifouling property of the hydrogel coating is influenced by its physical/chemical properties, such as water absorption property and weight loss rate (Su et al., 2020). Therefore, it is necessary to measure the water absorption property and weight loss rate of the hydrogel coating. The details in the following: the hydrogel coating prepared according to the 2.4 process was immersed in seawater, and 10 days was taken as an immersion cycle. Each cycle of immersion was reached, and the ABS coated with hydrogel coating was brought from seawater. Then, the wet weight (w_i) of samples was recorded after the surface was rinsed with absorbent filter paper to remove the excess water, and the dry weight (w_0) of the samples that were dried at 40 °C after 48 h was recorded. Each weight experiment was finished, and the sample was put into seawater to continue immersion.

The calculation about water absorption rate (S):

$$S = (w_i - w_0) / (w_i - w_0) \times 100\% \quad (1)$$

The calculation for the loss weight (D) was as follows:

$$D = 1 - (w_i - w_0) / (w_1 - w_0) \times 100\% \quad (2)$$

where w_0 is the weight of the ABS panel, w_s is the initial weight of ABS with hydrogel coating, w_i (i represents the number of soaking cycles and $i = 1, 2, 3, 4$ or 5) is the weight of the total ABS with hydrogel coating after soaking and drying, and w_t is the weight of total of ABS with hydrogel coating after absorbing water.

2.7. Antifouling experiments of the hydrogel coating

2.7.1. Anti-protein adsorption experiment

At room temperature, 20 mg bovine serum albumin (BSA) was

dissolved in 400 mL phosphate buffer solution (PBS, pH = 7.2) to obtain 0.05 mg/mL BSA solution. Thirty milliliters of PBS was placed into the Petri dish ($d = 10$ cm), and the hydrogel sample was prewetted for 2 h (Lin et al., 2017). Then, the PBS in the culture dish was poured out, 30 mL BSA solution was added, and the hydrogel sample was floated in BSA and immersed in the side with the hydrogel coating (Jeyachandran et al., 2009). After 24 h, the protein that did not adhere to the hydrogel coating surface was rinsed with 10 mL PBS to dislodge it. The PBS rinsing solution and BSA immersion solution were combined, and 20 mL PBS was added to the mixture, and left to stand for 2 min, and test solution A was obtained. The BSA content in test solution A was detected according to the Bradford (Kielkopf et al., 2020) to determine the anti-protein adhesion effect on the hydrogel coating. Briefly, the 1 mL protein solution to be tested was mixed with 10 mL Coomassie brilliant blue working solution. After 2 min, the absorbance of mixture solution was measured by UV spectrophotometry (HITACHI, U-2800, Japan) at 595 nm within 1 h. Then, the BSA concentration was calculated by a standard curve (Fig. S1).

The anti-protein adsorption rate (A) was calculated through the following formula, which is expressed by comparison with the BSA concentration:

$$A = C_1 / C_0 \times 100\% \quad (3)$$

where C_0 is the concentration of BSA solution that has not been used and C_1 is the protein concentration without adhesion on the coating. In the preparation of all solutions, bubbles must be avoided, otherwise the experimental results will be affected.

2.7.2. Algae adhesion test

Nitzschia closterium and Chlorella, which reached a growth index cycle, and nutritive salt-added seawater (nutritive salt composition and dosage are shown in Table 2) were mixed, and the concentration of the target algae solution was obtained by controlling the absorbance value of the algae solution to 0.09 at 680 nm by UV. The ABS panel with hydrogel coating was placed on the bottom of the weighing bottle (80×30 mm²), the hydrogel coating's side was upward, and then 40 mL of the above target algae solution was added. After experimental preparation, the weighing bottles were placed into an incubator with a light intensity of 4000 lux, a light-to-dark ratio of used time is 14:10, and a culture temperature of 21 °C. The culture time was 5 d, and the algae solution was shaken at a unified time every day during the 5 d. After 5 d of the experiment, the sample was removed from the weighing bottle, and the weakly adhesive algae on the hydrogel coating surface were

Table 2

Nutritive salt composition and dosage.

Algae	Nutritive salt(μ L/100 mL seawater)				
	Vitamin	sodium nitrate	sodium phosphate	sodium silicate	trace element
Chlorella	50	100	100		50
Nitzschia	50	100	100	100	50
Closterium					

Note: The composition and concentration of the nutrient salt solution are shown in the supplementary materials.

rinsed off. Then, all algae adhered on the hydrogel coating were thoroughly rinsed into another beaker with 40 mL nutritive salt solution, and the algae concentration was calculated by its absorbance and standard curve. The absorbance was tested by an ultraviolet spectrophotometer at 680 nm. The standard curve was prepared according to the microscopic counting method (Leica, CTR5000) and is shown in Fig. S1. The seawater used was funnel filtered and sterilized at 121 °C for 10 min by high temperature sterilization pot (MLS-3750, Japan).

2.7.3. Antibacterial assay

The marine bacteria (Marine Flavobacterium) were provided by The College of Marine Life Science of Ocean University of China. Marine bacteria were activated at 37 °C for 24 h and then cultured with 2216E liquid medium solution for 48 h. One hundred microliters of bacterial solution was added to the hydrogel surface, and the sample was sealed in a Petri dish (d = 10 cm), and then cultured it in an incubator at 37 °C. After 24 h, the bacterial solution on the hydrogel was rinsed with 10 mL SCDL solution, the experimental bacterial solution was obtained, and the experimental bacterial solution was diluted with normal saline. Then, the bacterial solution (100 µL) was evenly spread on the surface of the agar solid medium. When the bacterial solution was completely absorbed, it was sealed and placed in an incubator at 37 °C for 42 h. The above antibacterial experiments were carried out in a sterile environment, all liquids were autoclaved before use. The bacteriostatic rate was determined by Formula (4).

$$S = (S_0 - S_i) / S_0 \times 100\% \quad (4)$$

where S is the bacteriostatic rate, S_0 is the average number of colonies in the control group, and S_i is the average number of colonies of the experimental objects.

2.7.4. Experiment of hanging plates with hydrogel coatings in the actual marine environment

The marine antifouling performance of hydrogel coatings was tested in an actual marine environment refer to the "Shallow sea immersion test method for antifouling paint template" (GB/T 5370-2007). The substrate coated with the hydrogel coating is PVC plate (900 × 300 × 3 mm³), and the pretreatment process of the PVC plate was the same as ABS plate. The PVC plate with hydrogel coating was immersed to 1.5 m below sea level after drying. After dozens of days, the PVC plate with the hydrogel coating was gently washed with seawater and removed from the sea. Then the coating was completely and clearly photographed, and avoid the growth of fouling organisms on the hydrogel coating surface, it was returned to its original position as soon as possible. By observing the number and type of fouling organisms on the hydrogel coating, the marine antifouling property in the actual marine environment was judged. The experiment was carried out in August 2021 in the marine

area near Rongcheng Port, Shandong Province, China. The salinity of seawater is 29.00–32.00‰.

3. Results and discussion

3.1. Structural analysis

Because TPA has good antifouling performance and its marine antifouling performance has been proved by blending it into coatings physically (Cheng and Yu, 2010), it can also be used to prepare antifouling materials by chemical synthesis. Herein, a kind of TPA capsaicin derivative was synthesized by Friedel–Crafts reaction, whose chemical structure was confirmed by ¹H NMR in Fig. 2 (Zhang et al., 2022). ¹H NMR (DMSO-*d*₆, 600 MHz) δ: 9.17 (s, 2H, -OH), 8.72 (t, 1H, -NH-), 8.41 (s, 1H, -OH), 6.44 (s, 1H, -Ph), 6.25–6.15 (q, 4H, =CH₂), 5.64 (d, 2H, =CH-), 4.16 (d, 4H, -CH₂-).

Fig. 3a is the spectra of the prepolymers. It clearly shows the C=O stretching vibration peak of -COO- (BA) and -COOH (AA) at 1737 cm⁻¹, a peak at 1384 cm⁻¹ for the O-H bending vibration of the -COOH (AA), Ph-OH (O-H stretching vibration peak, TPA), the N-H of -CONH₂ (AM) stretch at 3733–3132 cm⁻¹ and the C=O of -CONH₂ (AM) stretch at 1672 cm⁻¹, and C-C of -Ph (TPA) stretch at 1500–1800 cm⁻¹, and the C-N stretching vibration peak of -CH₂N(CH₃)₂ (DM) and -CONH₂ (AM) at 1163 cm⁻¹. The above results demonstrated that these groups of monomers (AA, BA, AM, DM and TPA) existed in the spectra of the PTPA1, PTPA3, PTPA5 and PTPA7 prepolymers, and the target prepolymers were obtained. Fig. 3b shows the IR spectra of the hydrogel coatings of TPA1, TPA3, TPA5 and TPA7. Comparing these IR spectra to Fig. 3a, the intensities of some special group peaks are decreased, for example, the peak at 3364 cm⁻¹ of the -OH (-COOH) stretching vibration, the more intuitive comparison of PTPA7 and TPA7 in Fig. 3c. This contributes to aziridine reacting with -COOH to form the crosslinking structure -COO-CH₂-CH (CH₃)-NH-CH₂-, -OH (-COOH) was consumed. This demonstrates that the crosslinking structure of hydrogel polymer crosslinking with TMAP crosslinking agent, above result is also verified in the molecular weight distribution experiment. Fig. 4 is the molecular weight distribution graph of PTPA1 (non-crosslinked) and TPA1 (crosslinked), the number-average molecular weight of PTPA1 is 11976 g/mol, and the number-average molecular weight of TPA1 is 76716 g/mol, because aziridine reacting -COOH to form network structure and the number-average molecular weight is increased.

3.2. The adhesion properties of hydrogel coatings

The adhesive performance of hydrogel coating on the ABS plate was evaluated by static immersion in seawater (Fig. 5). After immersion for 50 d, none of the hydrogel coatings fell off of the substrate. This

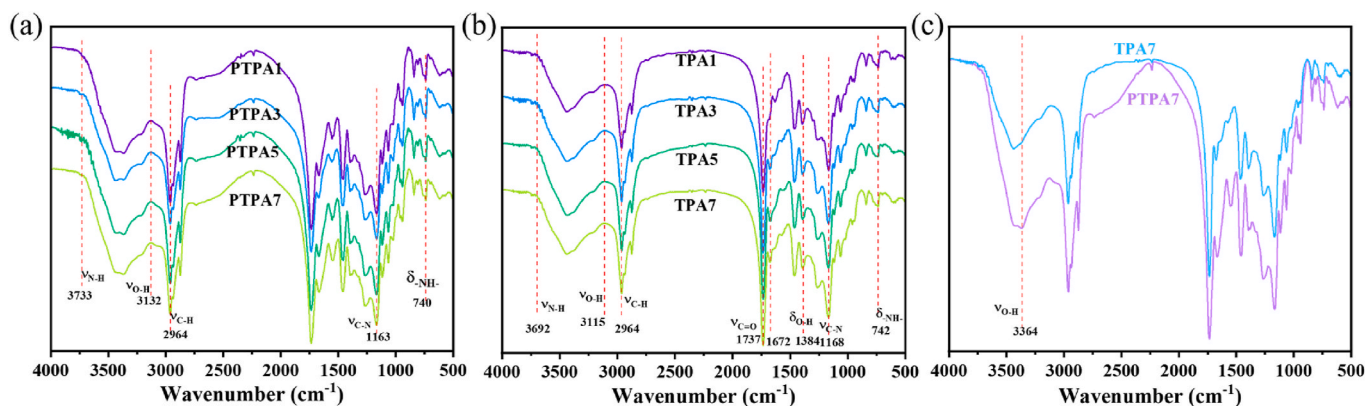


Fig. 3. The infrared spectroscopy spectra of hydrogels and their prepolymers. (a) Prepolymers: PTPA1, PTPA3, PTPA5 and PTPA7. (b) Hydrogels: TPA1, TPA3, TPA5 and TPA7. (c) PTPA7 and TPA7.

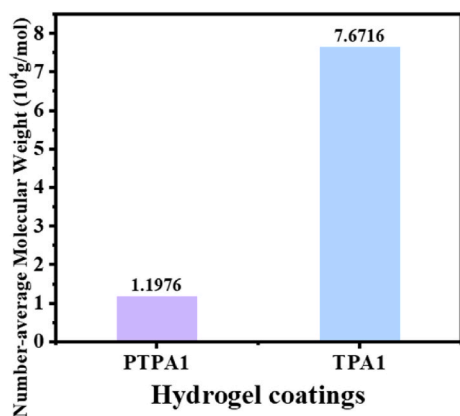


Fig. 4. Number-average molecular weight of PTPA1 and TPA1.

indicated that the hydrogel coatings had good adhesion ability and potential application. In generally speaking, the degradation rates of hydrogel coating were slow, more suitable for relatively long-term antifouling applications (Chen, 2022), hence, the weight loss rate of hydrogel coatings immersed for 50 d was observed (Fig. 6a), the experimental result indicated that all hydrogel coatings with TPA have a weight loss of less than 10%, and also proves that the hydrogel coating containing capsaicin derivatives has the potential for long-term application. Moreover, except TPA3, the weight loss rate of hydrogel coatings decreased with increasing capsaicin derivative content, because the molecular connectivity of the hydrogel would be increased by the crosslinking structure within TPA, the more the TPA content, the stronger the intermolecular connectivity and the less the hydrogel coating degradation, and also proves that the increase of capsaicin derivatives can enhance the potential of long-term antifouling application of hydrogel coatings.

3.3. The chemical and physical characterization of hydrogel coatings

The application of these hydrogel coatings is marine antifouling, and antifouling performance is usually related to their physical and chemical properties (Chen et al., 2022; Dai et al., 2021). Thus, the water-adsorption rate and contact angle of the hydrogel coatings are observed. The water-adsorption rate of hydrogel coatings increased with

the prolongation of the immersion time overall (Fig. 6b), this demonstrated that the hydrophilicity of hydrogel coatings increased with prolonged immersion time. The water-adsorption rate of hydrogel coatings decreased with the increasing content of TPA, and the contact angles increased with the increasing content of TPA, this showed that the hydrophilicity of the hydrogel coatings was reduced by the introduction of capsaicin derivative to a certain extent, because in general, the higher the water-adsorption rate was, the stronger the hydrophilicity and the smaller the contact angle (Fu et al., 2019; Su et al., 2020). Among all contact angles, the contact angle of TPA7 was increased most obviously, because there was a microphase separation structure in TPA7 according to Figs. 7 and 8, the surface contact angle would be increased by microphase separation structure (Alassod et al., 2022). The strong hydrophilicity and microphase separation contribute to the antifouling property of the hydrogel coating (Dong et al., 2022a; Zhao et al., 2013).

3.4. Antifouling properties

The initial antifouling property of the hydrogel coating can be evaluated by anti-protein adsorption. According to above analysis about the physical properties of hydrogel coatings, the water absorption rate was TPA1>TPA3>TPA5>TPA7, which demonstrated that the hydrophilicity is TPA1>TPA3>TPA5>TPA7. In response, the order of hydrogel coatings' anti-protein adsorption rate is TPA1>TPA3>TPA5>TPA7 (Fig. 9a). Hence, the protein adsorption resistance of the hydrogel coatings was reduced by the introduction of capsaicin derivative to a certain extent, but the protein adsorption resistance of all hydrogel coatings remains above 80%, because of the hydrophilicity of the hydrogel coatings, which proved that these coatings have certain antifouling ability.

The fluorescent coating has a repellent effect on fouling organisms such as diatoms (Bai et al., 2022; Cao et al., 2011; Hou et al., 2023). Fig. 8 shows that there is showed fluorescence on the surface of all hydrogel coatings containing TPA capsaicin derivatives, so this hydrogel coating has the repellent properties of fluorescent coatings. Moreover, the capsaicin derivative TPA is a repellent antifouling agent, and the number of diatoms that adhered to the surface of the coating decreased with increasing capsaicin derivative content (Fig. 9b). Thus, the antifouling performance of TPA can also be exerted when it was functionalized into polymers, and the diatom adhesive resistance of the hydrogel coatings was reduced by the introduction of the capsaicin derivative to a

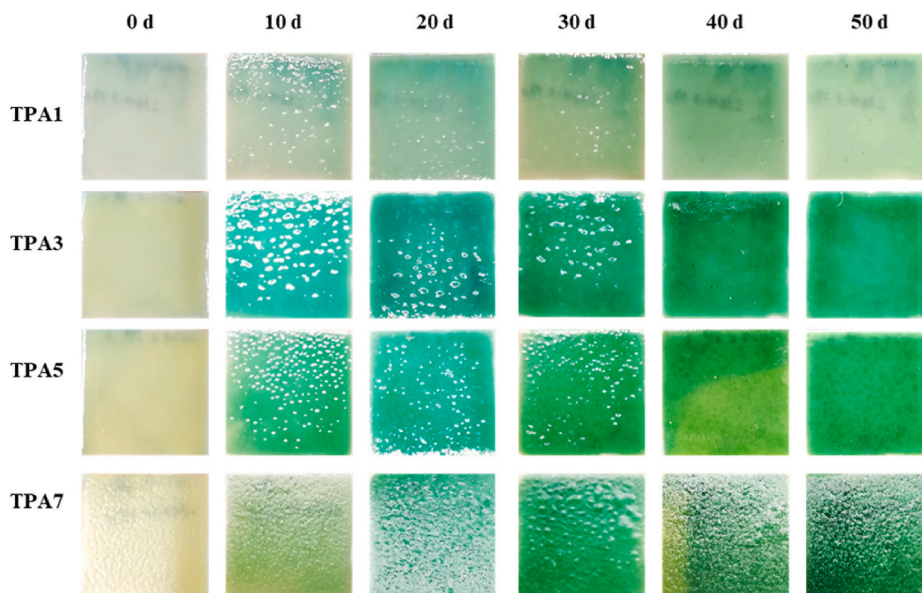


Fig. 5. The adhesion of TPA1, TPA3, TPA5 and TPA7 hydrogel coatings in static seawater immersion under different immersion cycles.

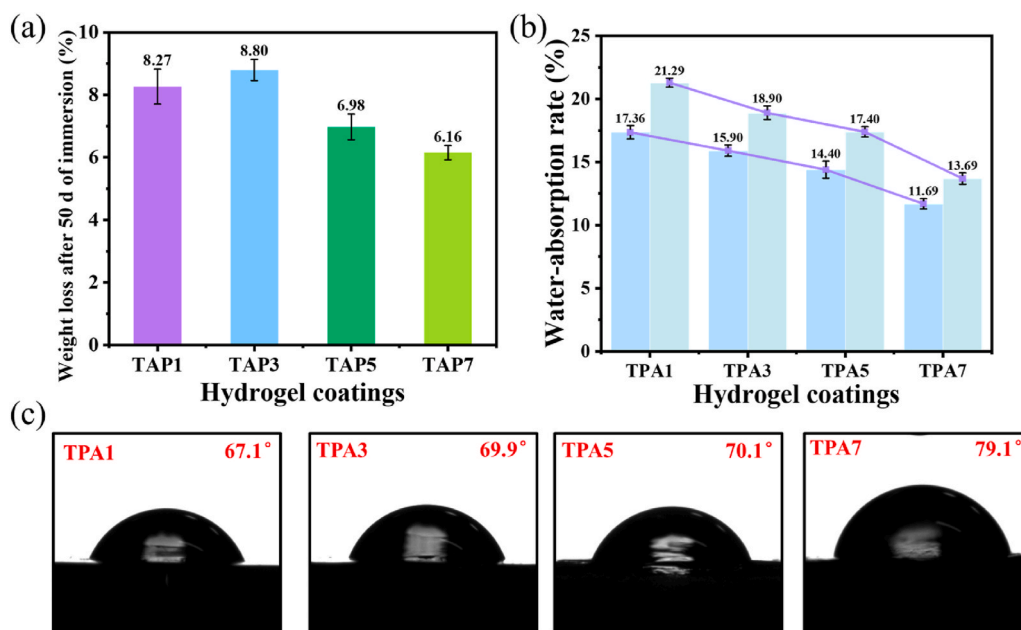


Fig. 6. (a) The weight loss rate of hydrogel coatings with 50 d of immersion (b) The change in water absorption of hydrogel coating with the change in soaking time. (c) The contact angle of hydrogel coating.

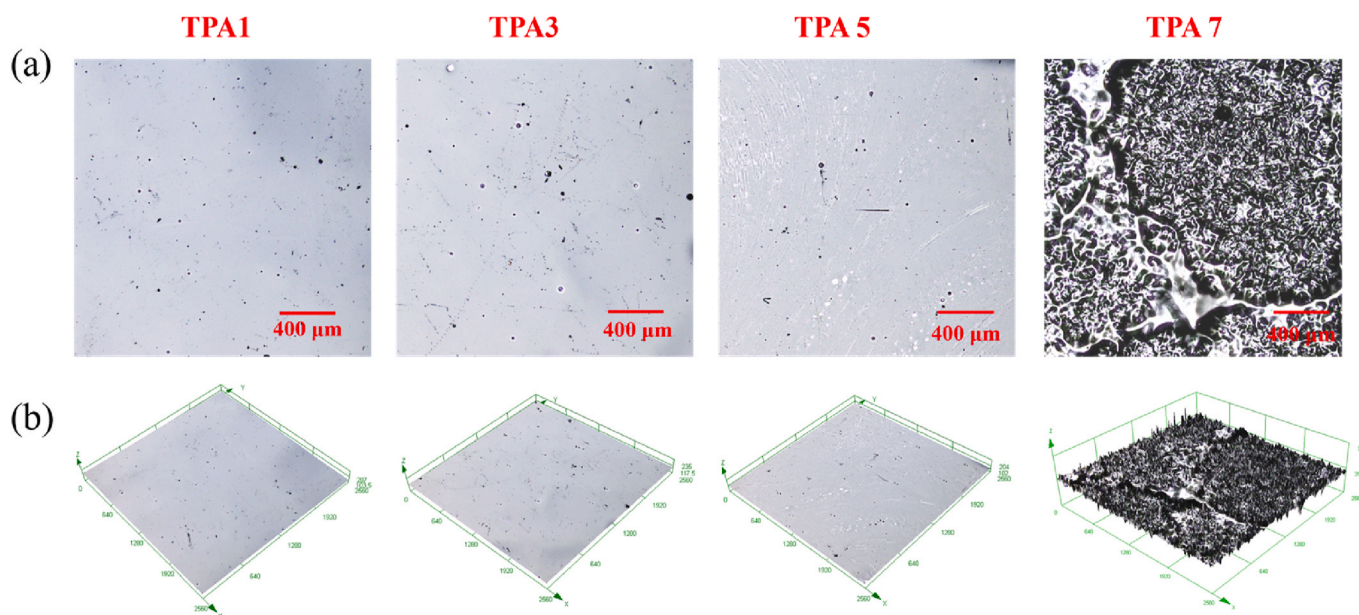


Fig. 7. Surface morphology of hydrogel coating observed by laser confocal microscopy. (a) 2D. (b) 3D.

certain extent. The maximum diatom density on the TPA1 surface was 1600 cells/mm², and the minimum diatom density on the TPA7 surface was only 431 cells/mm². What's more, the hydrogel coating has better algae anti-adhesion abilities than other hydrogel coatings. Fig. 10 shows the adhesion of *Chlorella* on the TPA7, CH and PVA surfaces. The number of *Chlorella* adhered on the surface of the TPA7 hydrogel coating was 438 cells/mm², which was far lower than that of the PVA hydrogel (2794 cells/mm²) according to Lin's work (Lin et al., 2019) and the CH hydrogel (3611 cells/mm²) according to Zhang's work (Zhang et al., 2022) about Cu₂O@CuO nanospheres-functionalized cellulose-based hydrogels. Therefore, the results also indicate that the hydrogel coatings containing TPA capsaicin derivative possesses reliable antialgae properties.

At the same time, the capsaicin derivative has antibacterial

properties, and quaternary ammonium salt DMC is also a well-known antifouling agent with antibacterial properties. The antibacterial rate of hydrogel coatings with different TPA capsaicin derivative contents and the antibacterial rate of hydrogels with the same DMC content were compared. It was found that the antibacterial rate of the hydrogel coating increased with increasing capsaicin derivative content obviously, and it can be clearly seen that the antibacterial rate of the DMC5 hydrogel coating with a DMC content of 5 mol% was lower than that of the hydrogel coatings with TPA capsaicin derivative contents of 3 mol% (TPA3), 5 mol% (TPA5) and 7 mol% (TPA7) (Fig. 11), which proved that the TPA derivatives can have a better antifouling effect than DMC in this hydrogel coating. Therefore, consistent with the results of algae inhibition experiments, the antifouling performance of the hydrogel coating can be improved significantly by the introduction of the TPA capsaicin

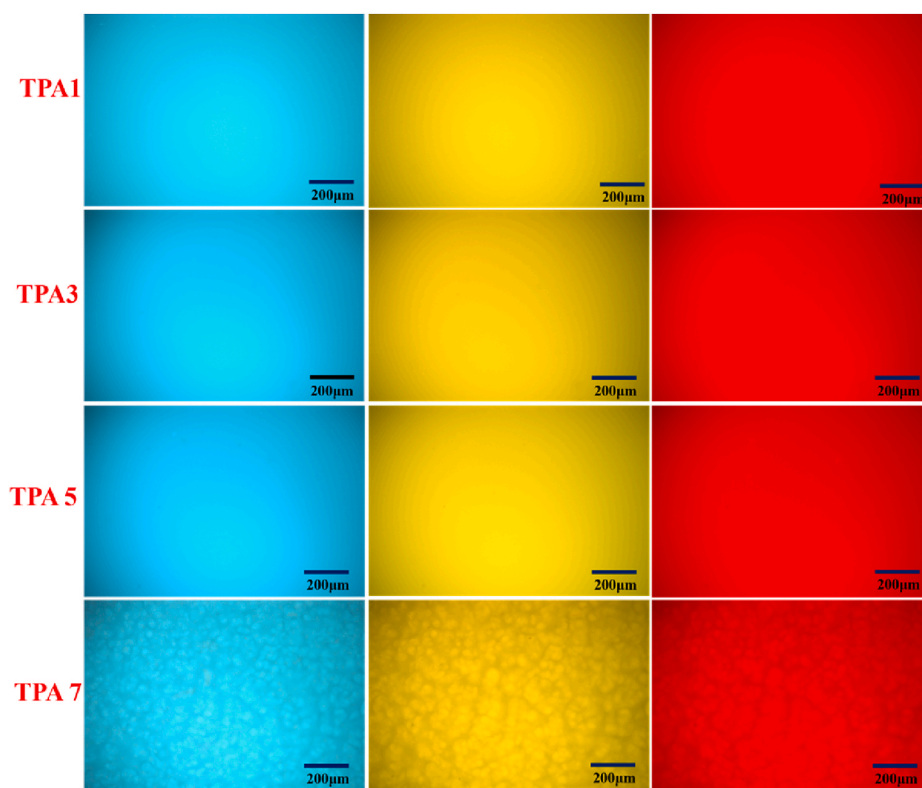


Fig. 8. Fluorescence imaging of hydrogel coating under three different excitation wavelengths.

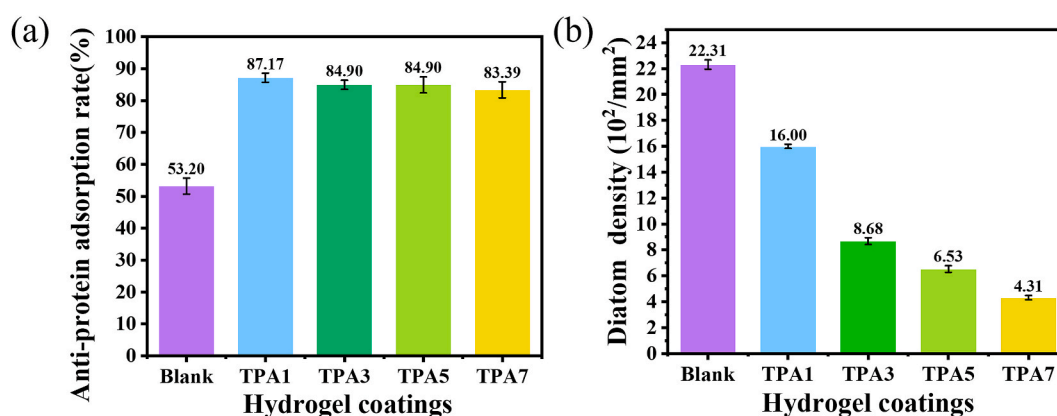


Fig. 9. Indoor antifouling test results. (a) Protein resistance of TPA1, TPA3, TPA5 and TPA7. (b) The *Nitzschia Closterium* density on the surface of TPA1, TPA3, TPA5 and TPA7 hydrogel coatings.

derivative.

The above indoor antifouling experimental results are also verified in the outdoor real marine antifouling test. The marine environment is complex and the fouling organisms are diverse, so testing the coating in the actual marine environment is the most effective way to evaluate the fouling organism prevention property of the coating. When the hydrogel coatings were experimented in the real marine environment for 73 d (Fig. 12 and Table S1), part of the DMC5 hydrogel coating fell off, and there is some black fouling dirt adhered on the surface. In the blank panel, the surface was adhered with a layer of biological mucosa and a small amount of mussels, while only a few aquatic plants adhered on the surface of the hydrogel coatings containing capsaicin derivative. In summary, the antifouling performance of hydrogel coatings can be improved by introducing the capsaicin derivative TPA, and the hydrogel coating modified by the capsaicin derivative TPA has the potential as a

marine antifouling coating.

The antifouling mechanism for hydrogel coatings containing capsaicin derivative is discussed in Fig. 13. First, the good hydrophilicity of the hydrogel coating could keep from the first stage of biofouling formation, inhibiting the adsorption of proteins. Second, the capsaicin derivative is a repellent antifouling agent, when the hydrogel coating is contacted by diatoms, bacteria and other microorganisms, the capsaicin derivative plays an antifouling role, and inhibits the adhesion of the microorganism. The reasons are as follows: (1) A capsaicin structure and an active factor exists in the capsaicin derivative, when the fouling organism contacts this compound, it will cause a spicy stimulation to the fouling organism, thereby the adhesion of fouling organism on the surface was inhibited. (2) Hydroxybenzene can entered the cell membrane, the membrane structure of fouling organism was damaged, the damaging effect can be promoted by the introduction of amide groups

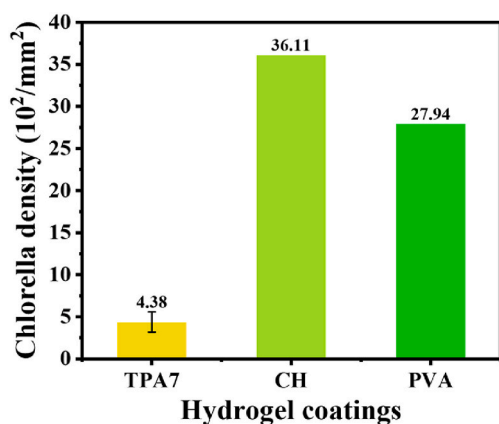


Fig. 10. The Chlorella density on the surface of TPA1, TPA3, TPA5 and TPA7 hydrogel coatings.

(Liu et al., 2016; Ni et al., 2018; Wang et al., 2020, 2021). Moreover, there are two antifouling mechanisms about microphase separation structure and fluorescent coating in these hydrogel coatings. Therefore, the hydrophilic property of the hydrogel coating and the antifouling performance of the capsaicin derivative work together to make the hydrogel coating a good antifouling performance.

4. Conclusion

Herein, a kind of antifouling hydrogel coating containing TPA

capsaicin derivative antifouling monomer was prepared by crosslinking BA, AA, AM, DM and TPA copolymer. The hydrophilicity, surface morphology, water absorption rate, contact angle and antifouling property of the hydrogel coatings were systematically assessed.

(1) The hydrogel coating has fluorescent properties and microphase separation structure. (2) All hydrogel coatings containing TPA capsaicin derivatives have certain anti-protein properties because of their hydrophilicity, and the anti-algae and antibacterial performance was also improved by used the repellent effect of TPA capsaicin derivatives on fouling organisms, and the antibacterial and anti-algae activity increased with increasing TPA content obviously. (3) Among them, TPA7 exhibited the best antibacterial efficiency against marine bacteria (reaching 95%), good anti-algae adsorption efficiency against *Nitzschia closterium* (algae density is only 431 cells/mm²) and *Chlorella* (438 cells/mm²), and a lower adhesive density of algae than other hydrogel coatings. (4) The hydrogel coating has obvious antifouling properties after 73 d of experiments in a real marine environment. Moreover, static seawater immersion panel experiment shows that there is application potential for the adhesion performance of the hydrogel coating.

This study provides a method for hydrogel coating to integrate other antifouling mechanisms to achieve broader antifouling effect, such coatings hold great future in marine antifouling and other related fields.

CRediT authorship contribution statement

Guangling He: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Yuhua Liu:** Conceptualization, Resources. **Wenyan Liu:** Data curation. **Lei Dong:** Validation. **Liangmin Yu:**

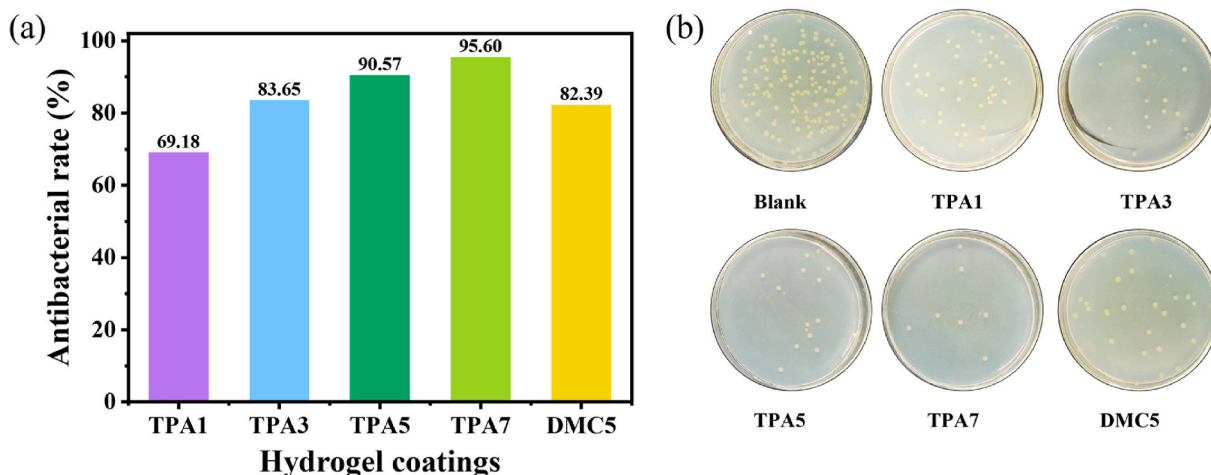


Fig. 11. The antibacterial effect and antibacterial rate image of capsaicin derivatives on marine bacteria.

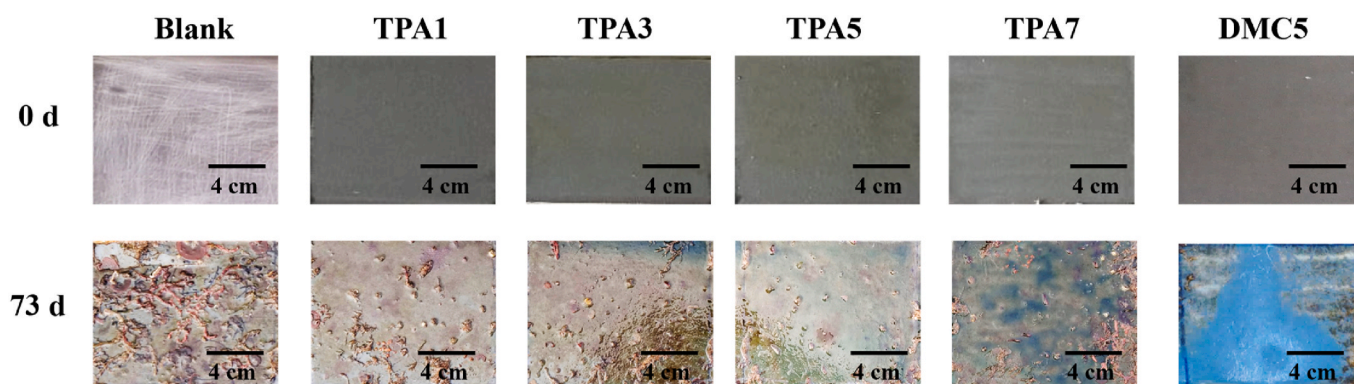


Fig. 12. In the actual marine environment, the biological adhesion condition of TPA1, TPA3, TPA5 and TPA7 hydrogel coatings changes with the experimental time.

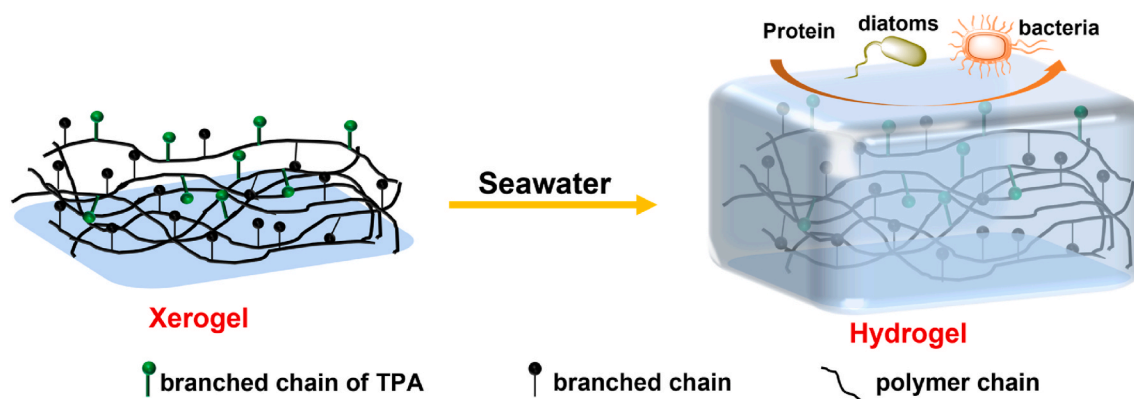


Fig. 13. Schematic diagram of antifouling mechanism of hydrogel coating after immersion in seawater.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jclepro.2023.139538>.

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